

● HARD

● ALGEBRA

Systems of two linear equations in two variables

01

10 questions · ~15 min

Q1

GRID-IN ALGEBRA

$$y = 4x + 1$$
$$4y = 15x - 8$$

The solution to the given system of equations is x, y . What is the value of $x - y$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$4x - 9y = 9y + 5$$

$$hy = 2 + 4x$$

In the given system of equations, h is a constant. If the system has no solution, what is the value of h ?

- A. -9
- B. 0
- C. 9
- D. 18

$$ax + by = 72$$

$$6x + 2by = 56$$

In the given system of equations, a and b are constants. The graphs of these equations in the xy -plane intersect at the point $(4, y)$. What is the value of a ?

- A. 3
- B. 4
- C. 6
- D. 14

Which system of linear equations has no solution?

A. $-2x + 3y = -9$
 $2x - 3y = 9$

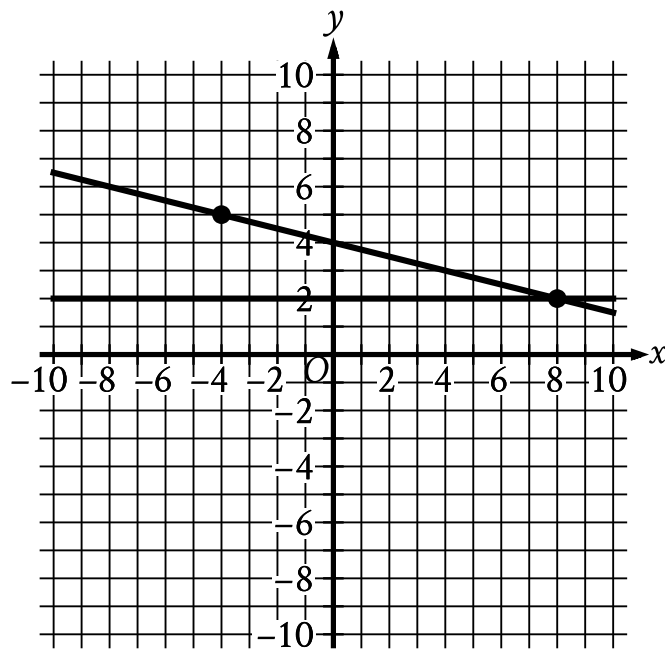
B. $2x - 3y = 9$
 $3x + 4y = 10$

C. $2x - 3y = 9$
 $-6x + 9y = -27$

D. $-2x + 3y = 9$
 $4x - 6y = 18$

Store A sells raspberries for \$ 5.50 per pint and blackberries for \$ 3.00 per pint. Store B sells raspberries for \$ 6.50 per pint and blackberries for \$ 8.00 per pint. A certain purchase of raspberries and blackberries would cost \$ 37.00 at Store A or \$ 66.00 at Store B. How many pints of blackberries are in this purchase?

- A. 4
- B. 5
- C. 8
- D. 12



- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (negative 4 comma 5)
 - (0 comma 4)
 - (8 comma 2)
- For the second line in the system:
 - The line is horizontal.
 - The line passes through the following points:
 - (0 comma 2)
 - (8 comma 2)

If a new graph of three linear equations is created using the system of equations shown and the equation $x + 4y = -16$, how many solutions x, y will the resulting system of three equations have?

A. Zero

- B.** Exactly one
- C.** Exactly two
- D.** Infinitely many

Q7

GRID-IN ALGEBRA

$$6 + 7r = pw$$

$$7r - 5w = 5w + 11$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q8

GRID-IN

ALGEBRA

$$x - 2 - 4y + 7 = 117$$

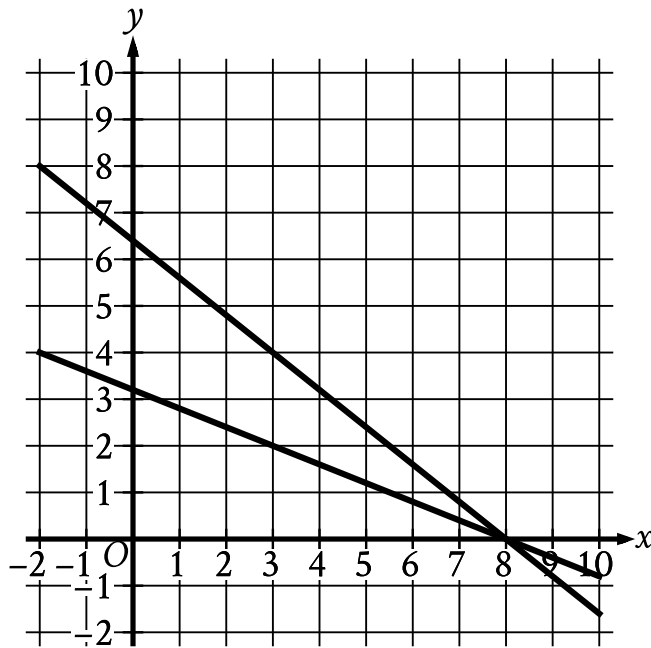
$$x - 2 + 4y + 7 = 442$$

The solution to the given system of equations is x, y . What is the value of $6x - 2$?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- For the first line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (3 comma 2)
 - (8 comma 0)
- For the second line in the system:
 - The line slants gradually down from left to right.
 - The line passes through the following points:
 - (3 comma 4)
 - (8 comma 0)

What system of linear equations is represented by the lines shown?

A. $8x + 4y = 32$
 $-10x - 4y = -64$

B. $8x - 4y = 32$

$$-10x + 4y = -64$$

C. $4x - 10y = 32$

$$-8x + 10y = -64$$

D. $4x + 10y = 32$

$$-8x - 10y = -64$$

$$48x - 72y = 30y + 24$$

$$ry = \frac{1}{6} - 16x$$

In the given system of equations, r is a constant. If the system has no solution, what is the value of r ?

STUDENT-PRODUCED RESPONSE

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☐ . ☐ / ☐ . ☐ / ☐ . ☐ .

☐ 0	☐ 0	☐ 0	☐ 0
☐ 1	☐ 1	☐ 1	☐ 1
☐ 2	☐ 2	☐ 2	☐ 2
☐ 3	☐ 3	☐ 3	☐ 3
☐ 4	☐ 4	☐ 4	☐ 4
☐ 5	☐ 5	☐ 5	☐ 5
☐ 6	☐ 6	☐ 6	☐ 6
☐ 7	☐ 7	☐ 7	☐ 7
☐ 8	☐ 8	☐ 8	☐ 8
☐ 9	☐ 9	☐ 9	☐ 9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.